

ResearchGraph AI — Abstract & System Overview

AI-Driven Academic Intelligence Platform integrating Vector Search, Knowledge Graphs, and LLMs

1. Abstract

ResearchGraph AI is an advanced academic research intelligence platform designed to transform literature review, scientific exploration, and hypothesis discovery by integrating Artificial Intelligence, Vector Search, Knowledge Graphs, and Large Language Models (LLMs) into a unified **GraphRAG (Graph-Augmented Retrieval Generation)** framework.

Traditional scientific search tools rely primarily on keyword matching or isolated vector similarity, which fail to capture complex inter-relations among papers, authors, algorithms, datasets, and research domains. ResearchGraph AI bridges this gap by harvesting open academic data from **OpenAlex** and **arXiv**, performing deep PDF extraction using **PyMuPDF**, generating semantic vector embeddings with **BAAI/BGE-large-en-v1.5** stored in **ChromaDB**, and constructing a rich multi-relational Knowledge Graph in **Neo4j**.

By fusing vector similarity retrieval with structural graph reasoning through **FastAPI** and **LangChain**, the platform enables citation-grounded natural language Q&A, automated literature review generation, multi-paper benchmarking, and automated research gap discovery. The user-facing interface, built with **React**, **TypeScript**, and **Tailwind CSS**, provides interactive visual analytics and intuitive graph exploration. Experimental validation demonstrates significant reductions in literature review duration, complete elimination of ungrounded LLM hallucinations, and enhanced domain contextual synthesis.

2. Limitations of Existing Systems

Traditional literature discovery tools suffer from fundamental architectural constraints:

- **Keyword Matching Bottlenecks:** Legacy academic databases (e.g., standard search portals) depend on keyword occurrences, failing to retrieve semantically relevant papers that express concepts using alternate terminology.
- **Isolated Vector RAG Limitations:** Standard vector search engines retrieve isolated text chunks based on cosine distance. However, they lack structural awareness regarding relationships between authors, datasets, baseline models, and methodological lineage.
- **Manual Synthesis Overhead:** Researchers spend dozens of hours manually scanning paper abstracts, extracting methodology details, and comparing benchmark results across published literature.
- **Generative Hallucination Risks:** Standard LLM chatbots often generate plausible-sounding but unverified citations or false claims due to lack of strict grounding in authoritative knowledge graphs and document repositories.

3. Advantages & Limitations Matrix

System Architecture	Key Advantages	Critical Limitations
Traditional Keyword Search (Google Scholar, Scopus)	• Rapid keyword lookup • Established citation counts	• Misses semantic intent • No automated synthesis • Manual relational linkage
Standard Vector RAG (Dense Embedding Search)	• Captures semantic similarity • Handles natural language	• Lacks graph relational reasoning • Chunk fragmentation • Ignores entity hierarchies
Proposed ResearchGraph AI (Hybrid GraphRAG)	• Unified Semantic & Graph Search • Verifiable citation answers • Automated Literature Review • Research Gap Identification	• Higher setup complexity • Requires dual-database indexing (ChromaDB + Neo4j)

4. How ResearchGraph AI Overcomes Existing Limitations

- **Hybrid GraphRAG Architecture:** Combines dense vector search (ChromaDB) with structured multi-hop graph querying (Neo4j) to supply LLMs with rich entity context and precise document chunks.
- **Verifiable Citation Grounding:** Enforces strict citation referencing using paper DOIs and metadata, guaranteeing zero-hallucination responses.
- **Automated Multi-Paper Comparison:** Automatically extracts datasets, algorithms, performance metrics, advantages, and limitations into structured comparison tables.
- **Automated Research Gap Identification:** Analyzes graph connectivity and paper limitations to highlight unaddressed research questions and emerging directions.